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Palladium-catalyzed directing group assisted and regioselectivity reversed cyclocarbonylation of arylallenes with 2-iodoanilines†

Jian-Shu Wang,‡[a] Lingyun Yao,‡[a] Jun Ying, *[a] Xiaoling Luo*[b] and Xiao-Feng Wu *[a,c]

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A palladium-catalyzed regioselective cyclocarbonylation of *N*-(2-pyridyl)sulfonyl (*N*-SO_2Py)-2-iodoanilines with allenes has been developed. With the assistance of the directing group *N*-SO_2Py, the regioselectivity of arylallenes was reversed and the reaction proceeded well to give various 3-methylene-2,3-dihydroquinolin-4(1*H*)-ones with up to 95% yield by using benzene-1,3,5-triyl triformate (TFBen) as the CO source. Control experiments and DFT calculations were performed to understand the reaction details and a possible reaction mechanism is proposed accordingly.

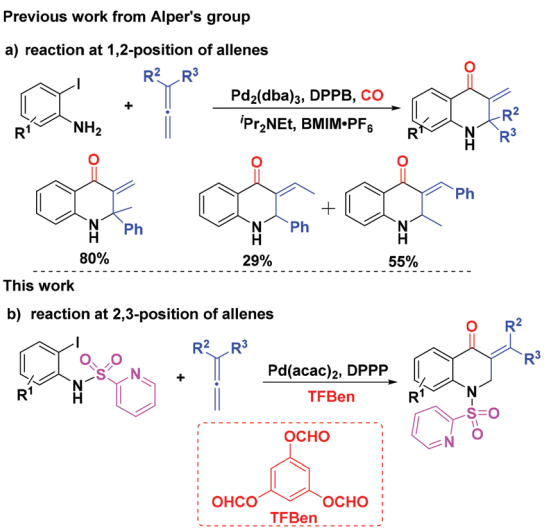
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Introduction

2,3-Dihydroquinolin-4(1*H*)-ones are considered as a type of crucial structural motif existing in many molecules with diverse biological activities such as analgesic,[1] anti-tumor,[2,3] and immunosuppressive properties.[4] In addition, they are important building blocks for the construction of valuable heterocyclic compounds.[5,6] Owing to their useful skeletons, a variety of synthetic methods to access 2,3-dihydroquinolin-4(1*H*)-ones have been realized.[7,8] Among all the 2,3-dihydroquinolin-4(1*H*)-ones, 3-methylene-2,3-dihydroquinolin-4(1*H*)-ones are a special class and usually have been prepared by the condensation of 2,3-dihydroquinolin-4(1*H*)-ones with aldehydes. Hence, new alternative methodologies for their preparation are always in demand.

Transition-metal-catalyzed carbonylative reactions have been recognized as one of the most straightforward and efficient methods to synthesize carbonyl-containing compounds.[9] Additionally, the chemistry of allenes in synthetic preparations is quite versatile.[10] Numerous novel synthetic procedures have been established. However, the reports on the carbonylative transformation with allenes are still limited compared with the reports on that with alkenes or alkynes.[11,12] The possible reasons are due to the high reactivity of allenes and the related regioselectivity issues.

Among the achievements, in 2007, Alper and Ye reported an attractive palladium-catalyzed ionic-liquid-promoted carbonylative cyclization of 2-iodoanilines with allenes.[12] Various 3-methylene-2,3-dihydroquinolin-4(1*H*)-ones were produced in moderate to good yields under 5 bar of CO (Scheme 1a).[12] This reaction occurred selectively at the 1,2-position of the alkyl substituted allenes applied. Under their conditions, buta-2,3-



Scheme 1 Carbonylative cyclization of 2-iodoanilines with allenes.

[a]*Department of Chemistry, Zhejiang Sci-Tech University, Xiasha Campus, Hangzhou 310018, People's Republic of China. E-mail: yingjun@zstu.edu.cn, xiao-feng.wu@catalysis.de*
[b]*Chongqing Key Laboratory of Inorganic Functional Materials, College of Chemistry, Chongqing Normal University, 401331 Chongqing, China. E-mail: xiaolingluo@cqnu.edu.cn*
[c]*Leibniz-Institut für Katalyse e. V. an der Universität Rostock, Albert-Einstein-Straße 29a, 18059 Rostock, Germany*

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‡These authors contributed equally to this work.

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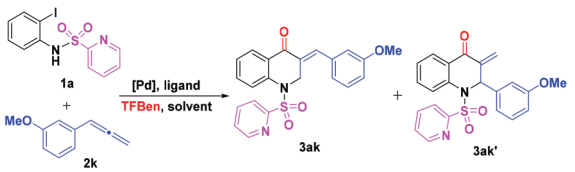
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dien-2-ylbenzene reacted only at the 1,2-position as well, while buta-1,2-dien-1-ylbenzene gave a mixture of two regioisomers. Herein, we report a new palladium-catalyzed cyclocarbonylation of (*N*-SO_2Py)-2-iodoanilines with allenes. With the assistance of the directing group, the regioselectivity was reversed and the reaction occurred only at the 2,3-position of the arylallenes applied. Various 3-methylene-2,3-dihydroquinolin-4(1*H*)-ones were produced in good yields with excellent regioselectivities by using TFBen[13] as the CO source (Scheme 1b).

Results and discussion

Initially, (*N*-SO_2Py)-2-iodoaniline **1a** and allene **2k** were selected as the model substrates with TFBen (4.0 equiv.) as the CO source in the presence of $Pd(OAc)_2$ (10 mol%), and various phosphine ligands were tested in dioxane at 90 °C for 24 h (Table 1, entries 1–5). Gratifyingly, using DPPP as the ligand with an appropriate bite angle, the reaction gave the cyclocarbonylated regiomers **3ak** and **3ak′** in 35% isolated yield with a ratio of 1 : 1 (Table 1, entry 4). Next, a series of solvents were investigated (Table 1, entries 6–9). It was found that employing MeCN as the solvent could enhance the yield (41%) and regioselectivity (5 : 1) (Table 1, entry 6). Notably, with the use of toluene, the target product **3ak** as a single regiomer was obtained in 53% yield (Table 1, entry 7). Subsequently, several palladium catalyst precursors were tested in this reaction (Table 1, entries 10–13). The reaction with $Pd(acac)_2$ improved the yield of **3ak** to 61% (Table 1, entry 12). Remarkably, the yield was increased to 87% when the reaction system was diluted (Table 1, entry 14). Additionally, 70% yield of the product **3ak** could be achieved even when using only 1 mol% of $Pd(acac)_2$ (Table 1, entry 15). It was noteworthy that the reaction yield was maintained (85%) when the amount of TFBen was reduced to 1.0 equivalent (Table 1, entry 16).

Table 1 Screening of the reaction conditions[a]



Entry	[Pd]	Ligand	Solvent	Yield (%) and ratio (**3ak** : **3ak′**)
1	$Pd(OAc)_2$	PPh_3	Dioxane	n.r.
2	$Pd(OAc)_2$	PCy_3	Dioxane	n.r.
3	$Pd(OAc)_2$	Xantphos	Dioxane	Trace
4	$Pd(OAc)_2$	DPPP	Dioxane	35 (1 : 1)
5	$Pd(OAc)_2$	DPPB	Dioxane	Trace
6	$Pd(OAc)_2$	DPPP	MeCN	41 (5 : 1)
7	$Pd(OAc)_2$	DPPP	Toluene	53 (**3ak**)
8	$Pd(OAc)_2$	DPPP	DMSO	Trace
9	$Pd(OAc)_2$	DPPP	DMF	Trace
10	$PdCl_2$	DPPP	Toluene	40 (**3ak**)
11	$Pd(TFA)_2$	DPPP	Toluene	55 (**3ak**)
12	$Pd(acac)_2$	DPPP	Toluene	61 (**3ak**)
13	$Pd(PPh_3)_2Cl_2$	DPPP	Toluene	34 (**3ak**)
14[b]	$Pd(acac)_2$	DPPP	Toluene	87 (**3ak**)
15[b,c]	$Pd(acac)_2$	DPPP	Toluene	70 (**3ak**)
16[b,d]	**$Pd(acac)_2$**	**DPPP**	**Toluene**	**85** (**3ak**)
17[b,e]	$Pd(acac)_2$	DPPP	Toluene	47 (**3ak**)

[a] Reaction conditions: **1a** (0.5 mmol), **2k** (2.5 equiv.), TFBen (4.0 equiv.), catalyst (10 mol%), monodentate ligand (20 mol%) or bidentate ligand (10 mol%), Et_3N (2.0 equiv.), solvent (1.0 mL), 90 °C, 24 h, isolated yield. [b] Toluene (2.5 mL). [c] $Pd(acac)_2$ (1 mol%). [d] TFBen (1.0 equiv.). [e] TFBen (0.5 equiv.).

With the optimal reaction conditions (Table 1, entry 16) in hand, we turned to investigate the scope of allenes **2** in the reaction with (*N*-SO_2Py)-2-iodoaniline **1a** (Scheme 2). The reaction with phenylallene **2a** gave the expected product **3aa** in 87% yield, whose structure was unambiguously confirmed by X-ray crystallography (CCDC 2023302, see the ESI†).[14]

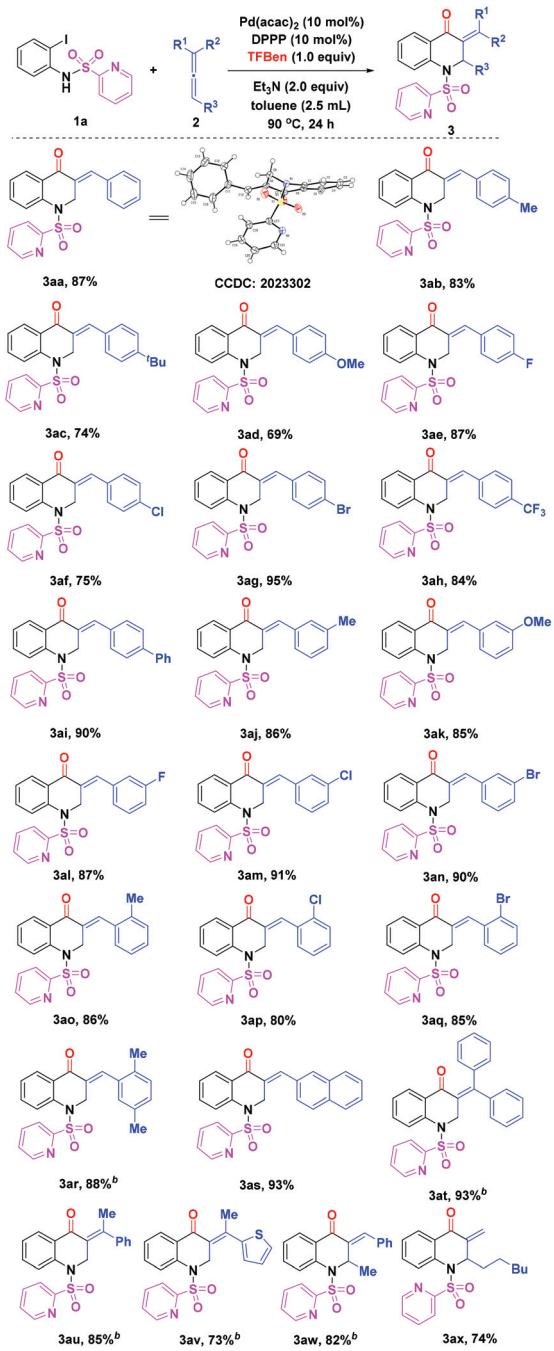
Phenylallenes bearing either *para*-electron-donating or *para*-electron-withdrawing substituents could undergo the reaction smoothly to give the corresponding products **3ab–ai** in high yields (69–95%). It was shown that the reaction with *meta*-substituted phenylallenes proceeded well to afford the desired products **3aj–an** in excellent yields (85–91%). Furthermore, good yields of the products **3ao–aq** were obtained when phenylallenes with *ortho*-substituents such as Me, Cl, and Br were tested. For 2,5-dimethyl phenylallene **2r**, the reaction led to the product **3ar** in 88% yield. Additionally, naphthylallene **2s** could be converted into the desired product **3as** in an excellent yield (93%). Notably, when 1,1- or 1,3-disubstituted allenes were subjected to the reaction system, a series of the target products **3at–aw** were achieved in good yields (73–93%). Aliphatic allene was tested as well, and 74% yield of the corresponding product **3ax** was obtained. However, the reaction occurred at the 1,2-position in this case.

Subsequently, a variety of (*N*-SO_2Py)-2-iodoanilines **1** were prepared and tested in the reaction with phenylallene **2a** (Scheme 3). For the substrates bearing functional groups such as Me, MeO, F, CF_3, COMe, COOMe, and Ph at the C4 position, the reaction afforded the corresponding products **3ba–ha** in high to excellent yields (80–93%). Moreover, the compounds having substituents such as Cl and Br at the C5 position were successfully converted into the desired products **3ia** and **3ja** in 88% and 93% yields, respectively. It was also found that the reaction with di-substituted compounds could lead to the products **3ka** and **3la** in high yields as well.

In order to show the utility of this protocol, a series of transformations of the product **3aa** were accomplished to furnish diverse 2,3-dihydroquinolin-4(1*H*)-ones with important functional moieties (Scheme 4).[15] The dibromination of **3aa** in the presence of *N*-bromosuccinimide and benzoic acid as the catalyst at 50 °C for 24 h afforded dibromo compound **4** in 90% yield (Scheme 4, eqn (1)). When **3aa** was treated with benzyl amine mediated by I_2/TBHP at room temperature, 75% yield of aziridine **5** was obtained (Scheme 4, eqn (2)). Additionally, **3aa** was transformed into dicarbonyl compound **6** in 87% yield *via* the thiazolium-catalyzed Stetter reaction (Scheme 4, eqn (3)).

To our delight, when compound **6** was treated with excess zinc powder in a mixed solvent (THF/$NH_4Cl_{(aq)}$ = 1 : 1), the removal of the *N*-SO_2Py directing group was realized in a facile

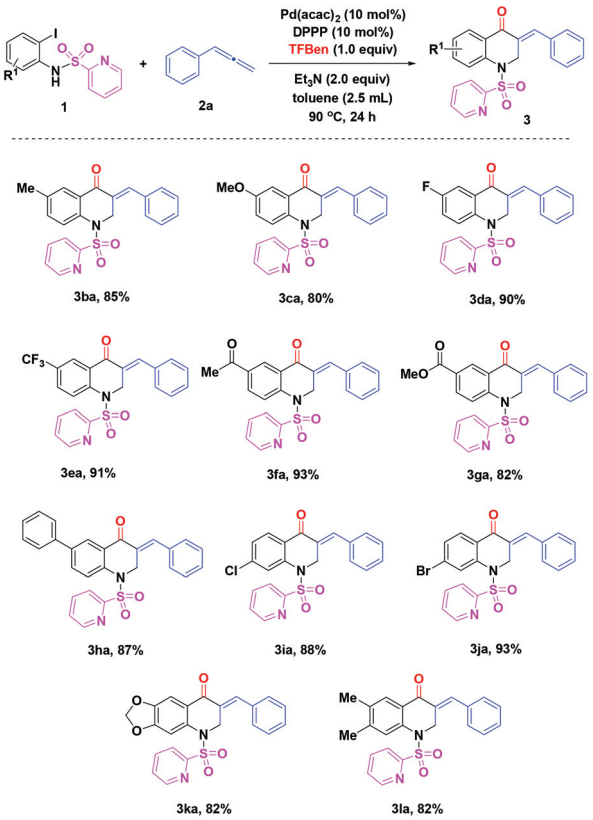
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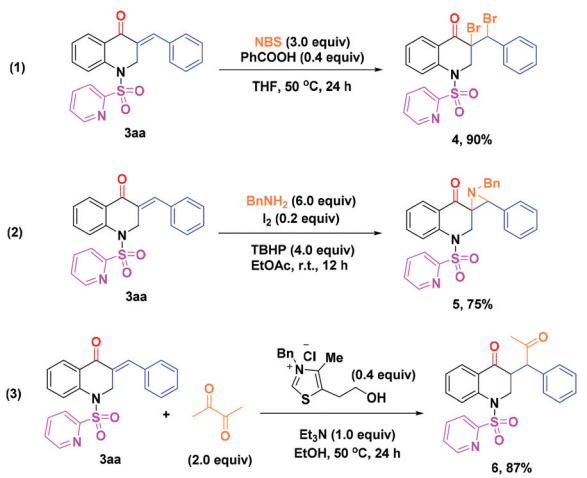
Scheme 2 Scope of allenes. Reaction conditions: **1a** (0.5 mmol), **2** (2.5 equiv.), TFBen (1.0 equiv.), $Pd(acac)_2$ (10 mol%), DPPP (10 mol%), Et_3N (2.0 equiv.), toluene (2.5 mL), 90 °C, 24 h, isolated yield. [a] 36 h.

manner to access the free amino product **7** in 78% yield, which could provide further derivatizations of 2,3-dihydroquinolin-4(1*H*)-one skeletons (Scheme 5).

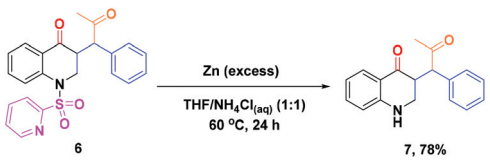
Next, we studied the effect of directing groups on 2-iodoanilines in the carbonylation with phenylallene **2a**, and the results are shown in Scheme 6. Initially, it was found that the reactions with unprotected 2-iodoaniline **8**, Me-2-iodoaniline **9**, and Boc-2-iodoaniline **10** gave complicated mixtures without



Scheme 3 Scope of (*N*-SO_2Py)-2-iodoanilines. Reaction conditions: **1** (0.5 mmol), **2a** (2.5 equiv.), TFBen (1.0 equiv.), $Pd(acac)_2$ (10 mol%), DPPP (10 mol%), Et_3N (2.0 equiv.), toluene (2.5 mL), 90 °C, 24 h, isolated yield.



Scheme 4 Transformations of the product **3aa**.



Scheme 5 Removal of the *N*-SO_2Py directing group.

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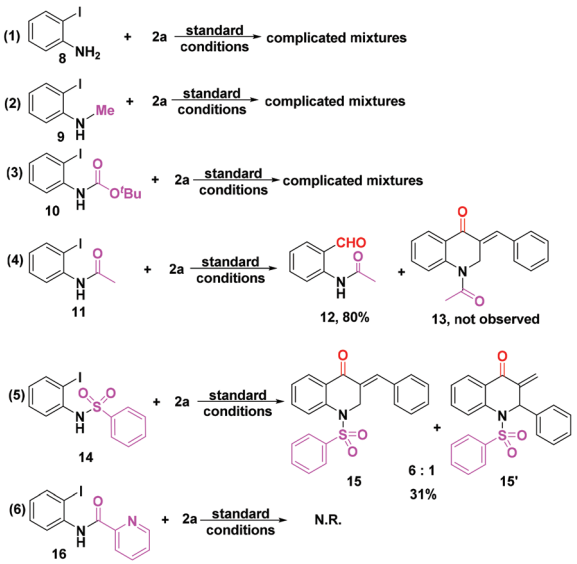
the expected cyclocarbonylated products (Scheme 6, eqn (1)–(3)). When Ac-2-iodoaniline **11** was subjected to the standard conditions, 80% of aldehyde **12** was obtained and the target product **13** was not observed (Scheme 6, eqn (4)). Interestingly, *N*-benzenesulfonyl-2-iodoaniline **14** could undergo the reaction to give the cyclocarbonylated regiomers **15** and **15′** with a significantly reduced yield (31%) and regioselectivity (6 : 1), which suggested that the coordination of the palladium catalyst with the N atom on *N*-SO_2Py of 2-iodoanilines was crucial in this reaction (Scheme 6, eqn (5)). Finally, the reaction did not occur when *N*-picolinyl-2-iodoaniline **16** was tested, indicating that the sulfuryl unit on the directing group also played an important role in promoting the cyclocarbonylation (Scheme 6, eqn (6)).

In order to have a better understanding of the reaction pathway, a possible reaction intermediate **17** was prepared and tested under our standard conditions (Scheme 7). The reaction of compound **17** with or without phenylallene **2a** under the standard reaction conditions produced intramolecular Heck cyclization product **18** in 80% or 30% yield, respectively. In both cases, the cyclocarbonylated product **3aa** was not observed. These findings showed that the carbonylation step was probably prior to the reaction of the NH group with allenes.

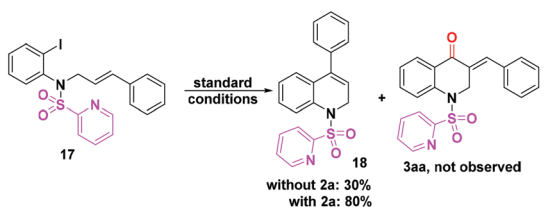
According to the experimental observation and previous theoretical studies[16] on the mechanism, two possible pathways for this palladium-catalyzed cyclocarbonylation of 2-iodoanilines with allenes are proposed (Scheme 8). The reaction starts from DPPP-coordinated Pd(0) species **A**, which is produced from $Pd(acac)_2$ with phosphine or CO as the reductant and then reacts with 2-iodoanilines by oxidative addition to afford aryl Pd(II) intermediate **B**. Next, CO insertion into the Pd–C (aryl) bond gives rise to acylpalladium intermediate **C** as the common initiation of the two pathways. In path I, allene migratory insertion occurs into the Pd–C(carbonyl) bond followed by intramolecular transformation to generate allylpalladium complex **D**, which can undergo deprotonation by the extra base NEt_3 to afford allylpalladium species **E**. Subsequent nucleophilic addition gives cyclocarbonylation product **3**. Alternatively, for path II, acylpalladium intermediate **C** can undergo deprotonation first to afford five-membered palladacycle **F**, followed by allene migratory insertion into the Pd–C (carbonyl) bond forming seven-membered palladacycle **G**. Reductive elimination provides the same product **3** finally.

To understand the mechanism and the regioselectivity of the palladium-catalyzed cyclocarbonylation of 2-iodoanilines with allenes, we performed a detailed DFT investigation. The calculated results of the favored path are shown in Fig. 1.

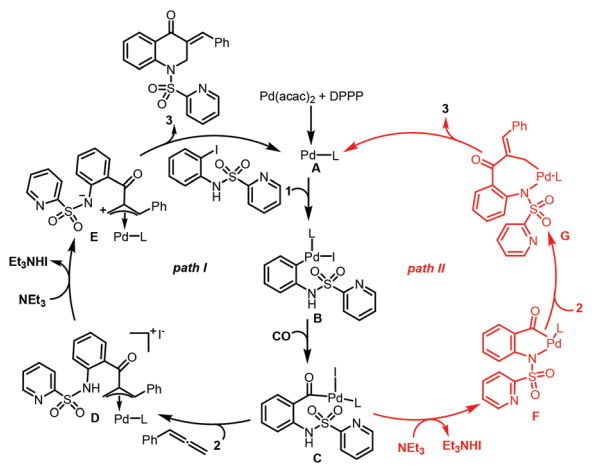
The catalytic cycle begins with the oxidative addition of the aryl iodide moiety of **1** to active Pd(0) catalyst **19** and generates coordinative intermediate **20**, which is exergonic by 8.6 kcal mol^{-1}. The energy of intermediate **20** is set as the zero point for the whole catalytic cycle. Oxidative addition of the aryl iodamide moiety occurs *via* three-membered-ring transition state **21-ts** to form aryl Pd(II) intermediate **22**, which is strongly exergonic by 29.0 kcal mol^{-1}. The activation free energy for this step is 10.4 kcal mol^{-1}. Then, CO replaces the iodide and coordinates with the central palladium atom to give species **23**, followed by CO insertion into the Pd–C(aryl) bond *via* three-membered-ring transition state **24-ts** to form carbonylpalladium compound **25**. This process is exergonic by 15.5 kcal mol^{-1} and the barrier is 13.5 kcal mol^{-1}. Subsequently, allene **2** replaces the iodide and coordinates with the central palladium atom to form π-coordinative inter-



Scheme 6 Effect of directing groups.



Scheme 7 Control experiments.



Scheme 8 Proposed mechanism.

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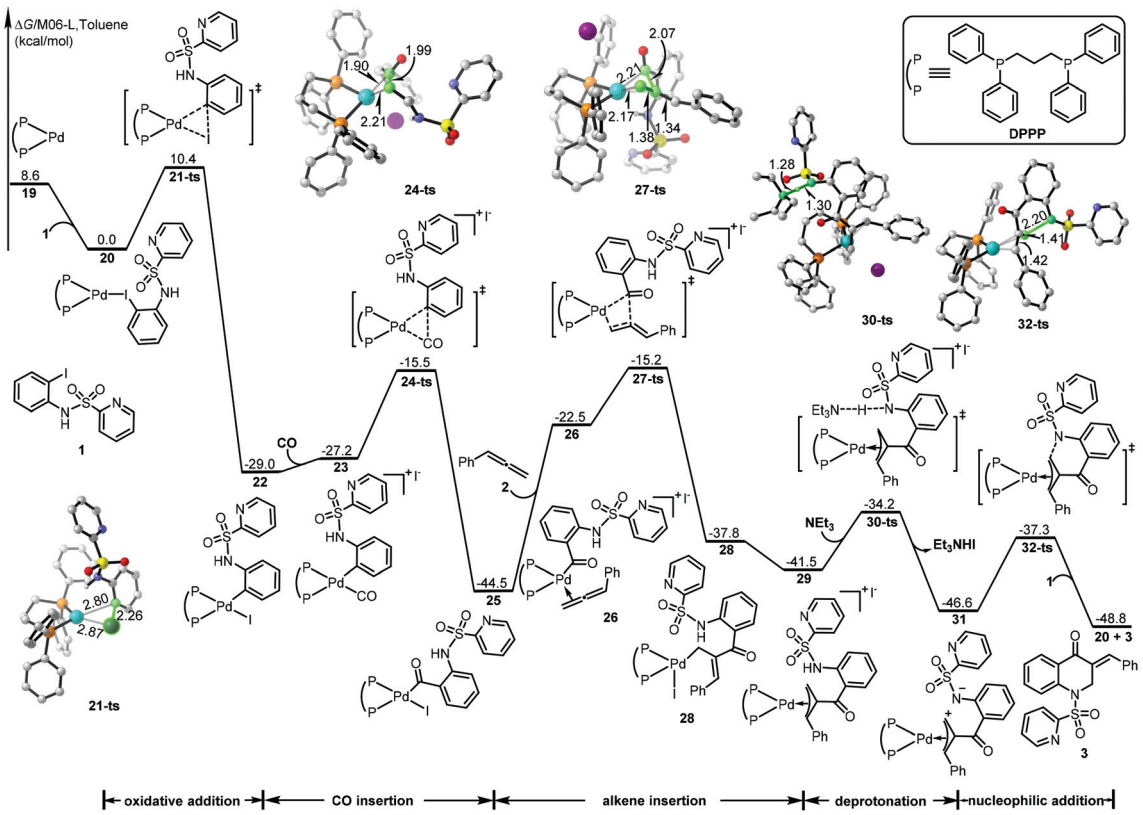


Fig. 1 Free energy profiles of palladium-catalyzed regioselective cyclocarbonylation of 2-iodoanilines with allenes. The energies are in kcal mol^{-1} and represent the relative free energies calculated with the DFT/M06-L method in toluene. The bond distances are in angstrom. The C–H bonds are omitted for clarity.

mediate **26**. Migratory insertion of the coordinated allene into the Pd–C(carbonyl) bond then occurs *via* transition state **27-ts** with an activation free energy of 29.3 kcal mol^{-1} to form alkylpalladium intermediate **28**. Allylpalladium complex **29** arises by intramolecular transformation with the dissociation of iodide. With the assistance of the base NEt_3, proton transfer occurs from the amino group to iodide *via* **30-ts** with a free energy barrier of only 10.3 kcal mol^{-1}. After deprotonation, zwitterionic allylpalladium intermediate **31** is generated with the dissociation of Et_3NHI. Finally, C–N bond formation with rapid nucleophilic addition *via* transition state **32-ts** generates cyclocarbonylation product **3** with the regeneration of the Pd (0) species. Coordination of another reactant **1** results in the regeneration of active species **20** to accomplish the full catalytic cycle. The rate determining step is the allene migratory insertion step with an activation free energy of 29.3 kcal mol^{-1}. Path II was calculated to be unfavored and the details are shown in Fig. S1.†

The regioselectivity was also considered for this reaction. As shown in Fig. 2, in the allene insertion process, allene 1,2-insertion into the Pd–C(carbonyl) bond can proceed *via* transition state **33-ts** to afford the corresponding byproduct **35**. The relative free energy of **33-ts** is 2.5 kcal mol^{-1} higher than that of **27-ts** relative to allene 2,3-insertion, which indicates that the 2,3-insertion product **3** is the main product thermo-

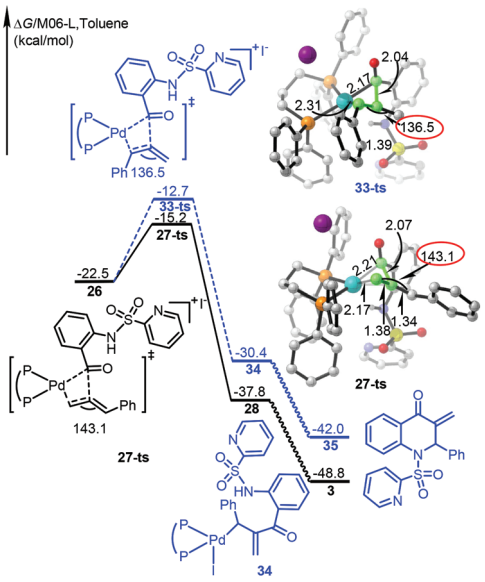


Fig. 2 Regioselectivity of the allene insertion step. The energies are in kcal mol^{-1} and represent the relative free energies calculated with the DFT/M06-L method in toluene. The bond distances are in angstrom. The C–H bonds are omitted for clarity.

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Table 2 Distortion energies[a] (ΔE_{dist}, kcal mol^{-1}) and interaction energies (ΔE_{int}, kcal mol^{-1}) of transition states **27-ts** and **33-ts**[b]

	$\Delta E_{dist(2)}$	$\Delta E_{dist(25)}$	ΔE_{dist}	ΔE_{int}	$\Delta E^{\ddagger}$
27-ts	19.0	47.3	66.3	−39.5	26.8
33-ts	24.5	47.7	72.2	−37.8	34.4

[a] The energies are calculated by using B3LYP/BS1 (sdd for Pd and I and 6-31g(d) for other atoms). [b] The activation energy of the transition state $\Delta E^{\ddagger}$ is composed of the energy required for the reactant to transform into the transition state (distortion energy, ΔE_{dist}) and the energy released after distortion to form the transition state (interaction energy, ΔE_{int}).

dynamically and kinetically. In order to illustrate the origin of regioselectivity, distortion–interaction energy analysis of the transition states **27-ts** and **33-ts** was performed and the calculated results are presented in Table 2. The distortion energy of the allene moiety is 19.0 kcal mol^{-1} in **27-ts** and 24.5 kcal mol^{-1} in **33-ts**, whereas the distortion energies of the carbonyl-palladium part and the interaction energies are similar. Correspondingly, the ∠C═C═C of the allene moiety in **27-ts** and **33-ts** is 143.1° and 136.5°, respectively. It confirms that more distortion of the allene moiety in **33-ts** results in a higher activation energy in the migratory insertion process.

Conclusions

In conclusion, we have developed a facile and efficient approach to access various 2,3-dihydroquinolin-4(1*H*)-ones in good yields and with excellent regioselectivities *via* a palladium-catalyzed cyclocarbonylation of (*N*-SO_2Py)-2-iodoanilines with allenes using TFBen as the CO source. This transformation can be realized successfully owing to the effect of the *N*-SO_2Py directing group. Control experiments and DFT calculations were performed to understand the reaction pathway.

Experimental

General procedure

(*N*-SO_2Py)-2-iodoaniline **1** (0.5 mmol, 1.0 equiv.), $Pd(acac)_2$ (0.05 mmol, 10 mol%), DPPP (0.05 mmol, 10 mol%), and a 2.5 mL vial containing TFBen (0.5 mmol, 1.0 equiv.) were added to an oven-dried tube (15 mL) which was then placed under vacuum and refilled with nitrogen three times. Allene **2** (1.25 mmol, 2.5 equiv.), Et_3N (1.0 mmol, 2.0 equiv.) and toluene (2.5 mL) were added into the tube *via* a syringe. The tube was sealed and stirred at 90 °C for 24 h. After the reaction was completed, the resulting mixture was concentrated under vacuum and purified by silica gel column chromatography (petroleum ether/ethyl acetate = 5 : 1) to obtain the product **3**.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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